

Node.js Developer Quiz (Medium–Hard Level)

Total Marks: 10

Instructions: Answer all questions. Marks are indicated against each question.

1. Explain the difference between the Node.js Event Loop phases and describe how `setImmediate()`, `process.nextTick()`, and `setTimeout()` are scheduled. (2 Marks)
2. Given a Node.js application experiencing memory leaks, describe at least three techniques or tools you would use to identify and troubleshoot the issue. (2 Marks)
3. What is the purpose of streams in Node.js? Compare Readable, Writable, Duplex, and Transform streams, and provide one practical use case for each. (2 Marks)
4. Consider an API that performs CPU-intensive image processing. Why might Node.js worker threads be preferable to the main event loop for this task? Explain the trade-offs. (2 Marks)
5. Write a short code snippet using `async/await` that fetches data from three independent APIs in parallel and handles errors gracefully. Briefly explain why your approach is efficient. (2 Marks)